## Appendix A. Search strategy used in the Chinese databases.

**Airiti Library:**

馬兜鈴酸 AND (尿路上皮癌 OR 尿路上皮腫瘤 OR 泌尿系腫瘤 OR 泌尿道上皮細胞癌 OR 移行上皮細胞癌 OR 腎盂癌 OR 膀胱 OR 輸尿管癌 OR 腎細胞癌)

**CNKI:**

馬兜鈴酸 並且包含 (尿路上皮癌 或者包含 尿路上皮腫瘤 或者包含 泌尿系腫瘤 或者包含 泌尿道上皮細胞癌 或者包含 移行上皮細胞癌 或者包含 腎盂癌 或者包含 膀胱癌 或者包含 輸尿管癌 或者包含 腎細胞癌)

**Wanfang:**

馬兜鈴酸 与 (尿路上皮癌 或 尿路上皮腫瘤 或 泌尿系腫瘤 或 泌尿道上皮細胞癌 或 移行上皮細胞癌 或 腎盂癌 或 膀胱癌 或 輸尿管癌 或 腎細胞癌)

**VIP:**

马兜铃酸 与 (尿路上皮癌 或 尿路上皮肿瘤 或 泌尿系肿瘤 或 泌尿道上皮细胞癌 或 移行上皮细胞癌 或 肾盂癌 或 膀胱癌 或 输尿管癌 或 肾细胞癌)

## Appendix B. Supplementary tables.

**Table S1.** Risk of bias within the cohort studies included.

| Author year | Selection | | | | Comparability | Outcome | | | Total quality score[b] |
|---|---|---|---|---|---|---|---|---|---|
| | Representativeness of exposed cohort | Selection of non-exposed cohort | Ascertainment of exposure[a] | Outcome present at start of study | Comparability of cohorts | Assessment of outcome | Length of follow-up | Adequacy of follow-up | |
| Li et al[50] 2007 | ★ | ★ | ★ | ★ | X | X | X | X | 4 |
| Zivcic-Cosic et al[37] 2007 | ★ | ★ | ★★ | ★ | X | X | X[C] | X | 5 |
| Li et al[38] 2008 | ★ | ★ | ★ | ★ | X | ★ | ★ | ★ | 7 |
| Wang et al[40] 2009 | ★ | ★ | ★★ | ★ | X | X | X | X | 5 |
| Zhou et al[41] 2009 | ★ | ★ | ★ | ★ | X | X | X | X | 4 |
| Xiao et al[42] 2010 | ★ | ★ | ★★ | ★ | X | X | ★ | ★ | 7 |
| Gao[43] 2011 | ★ | ★ | ★★ | ★ | X | X | X | X | 5 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Yang et al[45] 2011-primary UTC | ★ | ★ | ★★ | ★ | X | X | ★ | ★ | 7 |
| Yang et al[45] 2011 -recurrent UC | ★ | ★ | ★★ | ★ | X | X | ★ | ★ | 7 |
| Chen et al[51] 2013 | ★ | ★ | ★★ | ★ | X | ★ | ★ | ★ | 8 |
| Milenkovic-Petronic et al[52] 2014 | ★ | ★ | ★ | ★ | ★★ | ★ | ★ | ★ | 9 |
| Wang et al[47] 2014 | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | 8 |
| Liu et al[53] 2015 | ★ | ★ | ★★ | ★ | ★ | ★ | ★ | ★ | 9 |
| Chen et al[54] 2016 | ★ | ★ | ★★ | ★ | ★ | ★ | ★ | ★ | 9 |
| Ji et al[55] 2016 | ★ | ★ | ★ | ★ | X | ★ | ★ | ★ | 7 |
| Zhong et al[56] 2017 | ★ | ★ | ★ | ★ | ★★ | ★ | ★ | ★ | 9 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Wang et al[57] 2019 | ★ | ★ | ★ | ★ | ★ | ★ | ★ | ★ | 8 |
| Lu et al[58] 2020 | ★ | ★ | ★★ | ★ | X | ★ | ★ | ★ | 8 |
| Shan et al[59] 2020 | ★ | ★ | ★★ | ★ | ★ | ★ | ★ | ★ | 9 |
| **a.** Modified scoring criteria: a study which included patients diagnosed with AAN/BEN or AA-UC earned two stars. **b.** The total number of stars earned was presented as the total quality score (range 0–10). **c.** Not reported in the control group. X: no star earned. | | | | | | | | | |

**Table S2.** Risk of bias within the case-control studies included.

| **Author, year** | **Selection** | | | | **Comparability** | **Exposure** | | **Total quality score[b]** |
|---|---|---|---|---|---|---|---|---|
| | Adequacy of the case definition | Representativeness of the cases | Selection of controls | Definition of controls | Comparability of cases and cohorts | Ascertainment of exposure[a] | Non-response rate | |
| Sostaric et al[35] 1991 | ★ | ★ | ★ | ★ | X | ★ | ★ | 6 |
| Li et al[36] 2005 | ★ | ★ | ★ | ★ | X | ★★ | ★ | 7 |
| Lai et al[39] 2009 | ★ | ★ | ★ | ★ | ★ | ★ | ★ | 7 |
| Xiao et al[44] 2011 | ★ | ★ | ★ | ★ | X | ★ | ★ | 6 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Yang et al[46] 2013 | ★ | ★ | ★ | ★ | ★ | ★ | ★ | 7 |
| Hoang et al[48] 2016 | ★ | ★ | ★ | ★ | ★★ | ★ | ★ | 8 |
| Matic et al[49] 2016 | ★ | ★ | ★ | ★ | X | ★ | ★ | 6 |
| **a.** Modified scoring criteria: a study which included patients diagnosed with AAN/BEN or AA-UC earned two stars. **b.** The total number of stars earned was presented as the total quality score (range 0–9). X: no star earned. | | | | | | | | |